

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems.

(BL3) Assessment Tool Weightage: 20%

QUESTIONS: 20 Total Marks:20 Annexure A

Mock Interview

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

Annexure A

Mock Interview

1. What is Eigenfaces?
2. Define PCA in image processing.
3. What is contour-based shape representation?
4. What is region-based shape representation?
5. How does an active contour (snake) work?
6. How does a level set method work?
7. What is the Hough Transform?
8. How are Fourier descriptors used in shape analysis?
9. How are wavelet descriptors used in shape analysis?
10. What is a particle filter?
11. How is foreground–background separation done in surveillance systems?
12. How do multi-camera systems track pedestrians?
13. What is a Convolutional Neural Network (CNN)?
14. How does a Vision Transformer (ViT) work?
15. What is a Generative Adversarial Network (GAN)?
16. How is a Support Vector Machine (SVM) used for image classification?
17. How are Hidden Markov Models (HMM) applied in activity recognition?
18. What is a Gaussian Mixture Model (GMM)?
19. How is graph-based learning applied to 3D shape correspondence?

20. How is road and lane detection performed in autonomous driving?

1. What is Eigenfaces?

Eigenfaces is a face-recognition technique based on Principal Component Analysis (PCA). A set of training face images is converted into vectors, and PCA finds the most important variations among them. These principal components are called Eigenfaces. A new face is projected onto the Eigenface space and compared with stored face representations to identify the person.

Example: A face image can be recognized by comparing its Eigenface features with previously stored faces.

2. Define PCA in image processing.

Principal Component Analysis (PCA) is a dimensionality-reduction technique that transforms a large set of correlated image features into a smaller set of uncorrelated principal components.

The components are arranged according to the amount of variance they represent. The first component contains the maximum variance, the second contains the next highest variance, and so on.

Applications: Face recognition, image compression, feature extraction and noise reduction.

3. What is contour-based shape representation?

Contour-based shape representation describes an object using its boundary or outline rather than its internal region.

The boundary points are extracted from a binary or segmented image and used to represent the object's shape.

Examples: Chain codes, polygonal approximation and Fourier descriptors.

Application: Recognizing objects based on their external shape.

4. What is region-based shape representation?

Region-based shape representation represents an object using the entire region inside its boundary, including its area and pixel distribution.

It considers both the object's boundary and its internal structure.

Examples: Region moments, geometric moments and Zernike moments.

Difference: Contour-based methods use mainly the boundary, whereas region-based methods use the complete object region.

5. How does an active contour (snake) work?

An active contour or snake is a deformable curve that moves toward an object's boundary.

Initially, a curve is placed near the object. The curve is then iteratively modified using forces such as:

- **Internal forces** – maintain smoothness and continuity.
- **External image forces** – attract the curve toward edges.
- **Constraint forces** – guide the curve according to application requirements.

The process continues until the contour reaches a stable position around the object.

Application: Medical image segmentation, such as detecting organs or tumors.

6. How does a level set method work?

A level set method represents a moving contour implicitly using a higher-dimensional function called a level-set function.

Instead of directly moving boundary points, the function is updated according to image information. The zero-level contour represents the object's boundary.

It can naturally handle splitting and merging of contours, making it useful for complex shapes.

Application: Image segmentation, medical imaging and object tracking.

7. What is the Hough Transform?

The Hough Transform is a feature-detection technique used to detect geometric shapes such as lines and circles in an image.

For line detection, edge points are transformed into a parameter space. Points belonging to the same line produce a strong peak in this space, allowing the line to be detected.

Application: Road-lane detection, document analysis and shape detection.

8. How are Fourier descriptors used in shape analysis?

Fourier descriptors represent an object's contour using the Fourier transform.

The boundary coordinates are treated as a sequence and transformed into frequency components. The resulting coefficients describe the object's shape.

Low-frequency components represent the general shape, while high-frequency components represent fine details.

Advantages: They can provide compact shape representations and can be made relatively invariant to translation, rotation and scaling.

Application: Character and object recognition.

9. How are wavelet descriptors used in shape analysis?

Wavelet descriptors represent shape information at different scales or resolutions.

Wavelet decomposition separates the contour or shape signal into low-frequency and high-frequency components. Low-frequency components describe the overall shape, while high-frequency components capture fine details and local changes.

Advantage: They are useful for analyzing shapes containing both global and local features.

Application: Object recognition, texture analysis and industrial inspection.

10. What is a particle filter?

A particle filter is a probabilistic tracking algorithm used to estimate the state of a moving object when measurements are noisy or uncertain.

It represents the possible states using a set of particles, where each particle has a weight indicating its probability.

Main steps are:

1. Initialize particles.
2. Predict their new positions.
3. Compare predictions with the current observation.
4. Assign weights.
5. Resample particles based on their weights.

Application: Object tracking, pedestrian tracking and vehicle tracking.

11. How is foreground–background separation done in surveillance systems?

Foreground-background separation identifies moving objects as foreground and static scene areas as background.

A common method is background subtraction:

1. Build or estimate a background model.
2. Capture a new video frame.
3. Compare the frame with the background.
4. Calculate the difference.
5. Apply a threshold to obtain a foreground mask.
6. Apply morphological operations to remove noise.
7. Detect and track the foreground objects.

Example: In CCTV, a person entering an empty corridor is detected as foreground.

12. How do multi-camera systems track pedestrians?

Multi-camera pedestrian tracking combines information from multiple cameras to follow a person across different camera views.

The system generally performs:

1. Pedestrian detection in each camera.
2. Feature extraction such as appearance, color and shape.
3. Tracking within each camera.
4. Matching the same person across different cameras.
5. Combining the tracks into a continuous trajectory.

This is called multi-camera or cross-camera tracking.

Application: Airports, shopping malls, railway stations and smart surveillance.

13. What is a Convolutional Neural Network (CNN)?

A Convolutional Neural Network (CNN) is a deep learning model designed mainly for processing images.

A typical CNN contains:

- **Convolution layers** – extract features such as edges, textures and shapes.
- **Activation functions** – introduce non-linearity.
- **Pooling layers** – reduce spatial dimensions.
- **Fully connected/output layers** – perform classification.

Early layers learn simple features, while deeper layers learn complex features.

Applications: Image classification, face recognition, object detection and medical image analysis.

14. How does a Vision Transformer (ViT) work?

A Vision Transformer (ViT) applies the Transformer architecture to images.

The basic process is:

1. Divide the image into small patches.
2. Convert each patch into a numerical vector called an embedding.
3. Add positional information to preserve patch locations.
4. Pass the sequence of patch embeddings through Transformer encoder layers.
5. Use self-attention to learn relationships between different image regions.
6. Use the output for image classification or other vision tasks.

Key idea: Unlike CNNs, ViTs can directly model relationships between distant image regions using self-attention.

15. What is a Generative Adversarial Network (GAN)?

A Generative Adversarial Network (GAN) is a deep learning model used to generate new data that resembles a training dataset.

It consists of two networks:

- **Generator:** Creates fake samples.
- **Discriminator:** Determines whether a sample is real or generated.

The generator tries to fool the discriminator, while the discriminator tries to correctly distinguish real and fake samples. Through this competition, the generator learns to create realistic data.

Applications: Image generation, image enhancement, data augmentation and image-to-image translation.

16. How is a Support Vector Machine (SVM) used for image classification?

An **SVM** is a supervised machine-learning algorithm that classifies data by finding an optimal **decision boundary**, called a hyperplane.

For image classification:

1. Extract features from images.
2. Represent each image as a feature vector.
3. Train the SVM using labeled images.
4. Find the decision boundary that separates classes.
5. Use the trained SVM to classify new images.

An SVM with a kernel, such as an RBF kernel, can handle nonlinear classification problems.

Example: Classifying images as cat/dog or normal/abnormal medical images.

17. How are Hidden Markov Models (HMM) applied in activity recognition?

An HMM models activities as sequences of hidden states that generate observable data.

For activity recognition:

- **Hidden states:** Activities or stages of an activity.
- **Observations:** Features obtained from video, sensors or motion data.
- **Transition probabilities:** Probability of moving from one activity state to another.
- **Emission probabilities:** Probability of observing certain features for a state.

The observed sequence is given to the HMM, and algorithms such as Viterbi can determine the most likely sequence of hidden states.

Example: Recognizing activities such as walking, running, sitting and standing.

18. What is a Gaussian Mixture Model (GMM)?

A Gaussian Mixture Model (GMM) represents data as a combination of several Gaussian probability distributions.

Each Gaussian component has:

- Mean
- Covariance

- Mixing weight

The probability of a data point is calculated as a weighted combination of these Gaussian distributions.

GMMs are often trained using the Expectation-Maximization (EM) algorithm.

Application in computer vision: Background modeling, image segmentation, object detection and clustering.

19. How is graph-based learning applied to 3D shape correspondence?

In 3D shape correspondence, different 3D objects are represented using graphs.

- Nodes represent points, vertices or local regions of the 3D shape.
- Edges represent relationships between neighboring points.
- Node and edge features describe geometric properties.

Graph-based learning then learns relationships between two 3D shapes and identifies corresponding points or regions.

Example: Finding corresponding anatomical points between two 3D human body scans.

Advantage: It can capture both local geometric features and relationships between neighboring points.

20. How is road and lane detection performed in autonomous driving?

Road and lane detection identifies the drivable road area and lane boundaries from camera or sensor data.

A typical process is:

1. Capture a frame from the front-facing camera.
2. Convert and preprocess the image.
3. Apply noise reduction.
4. Detect edges using Canny edge detection.
5. Select the road region using ROI.
6. Detect lane lines using Hough Transform or a learned lane-detection model.
7. Estimate curved lanes when necessary.
8. Track lane positions across consecutive frames.
9. Handle shadows, lighting changes and partial occlusion.

10. Overlay the detected lanes on the original image.